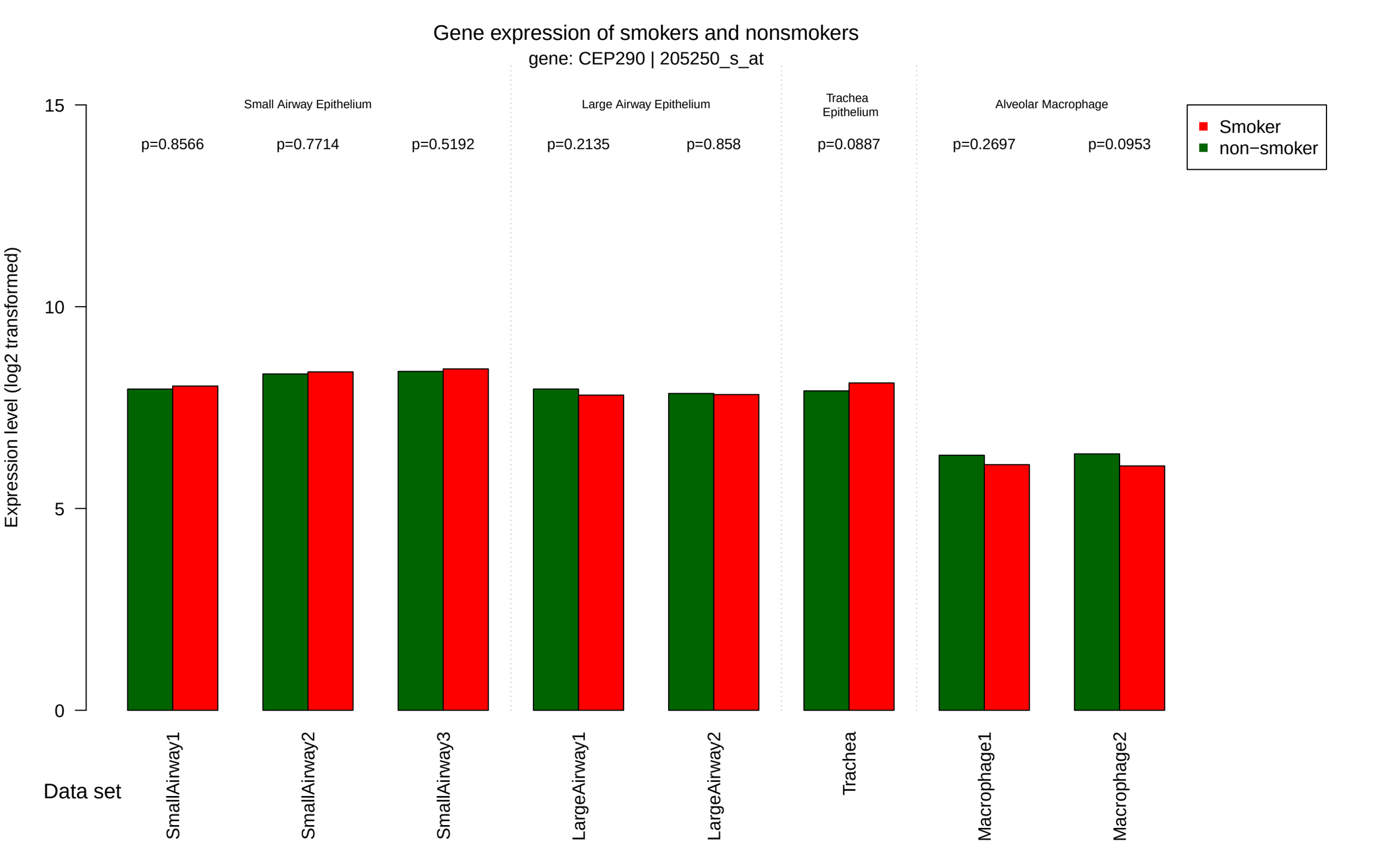
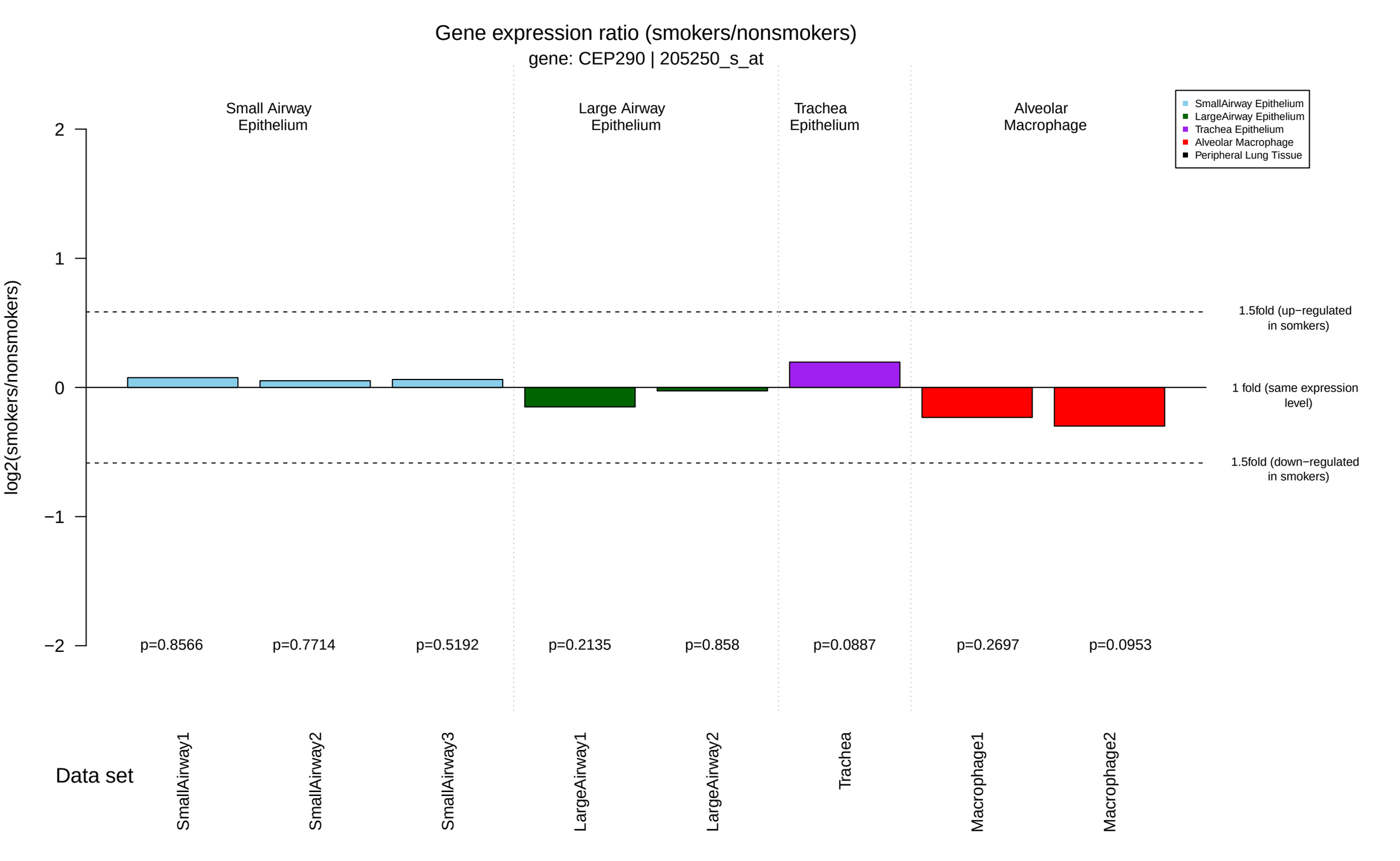
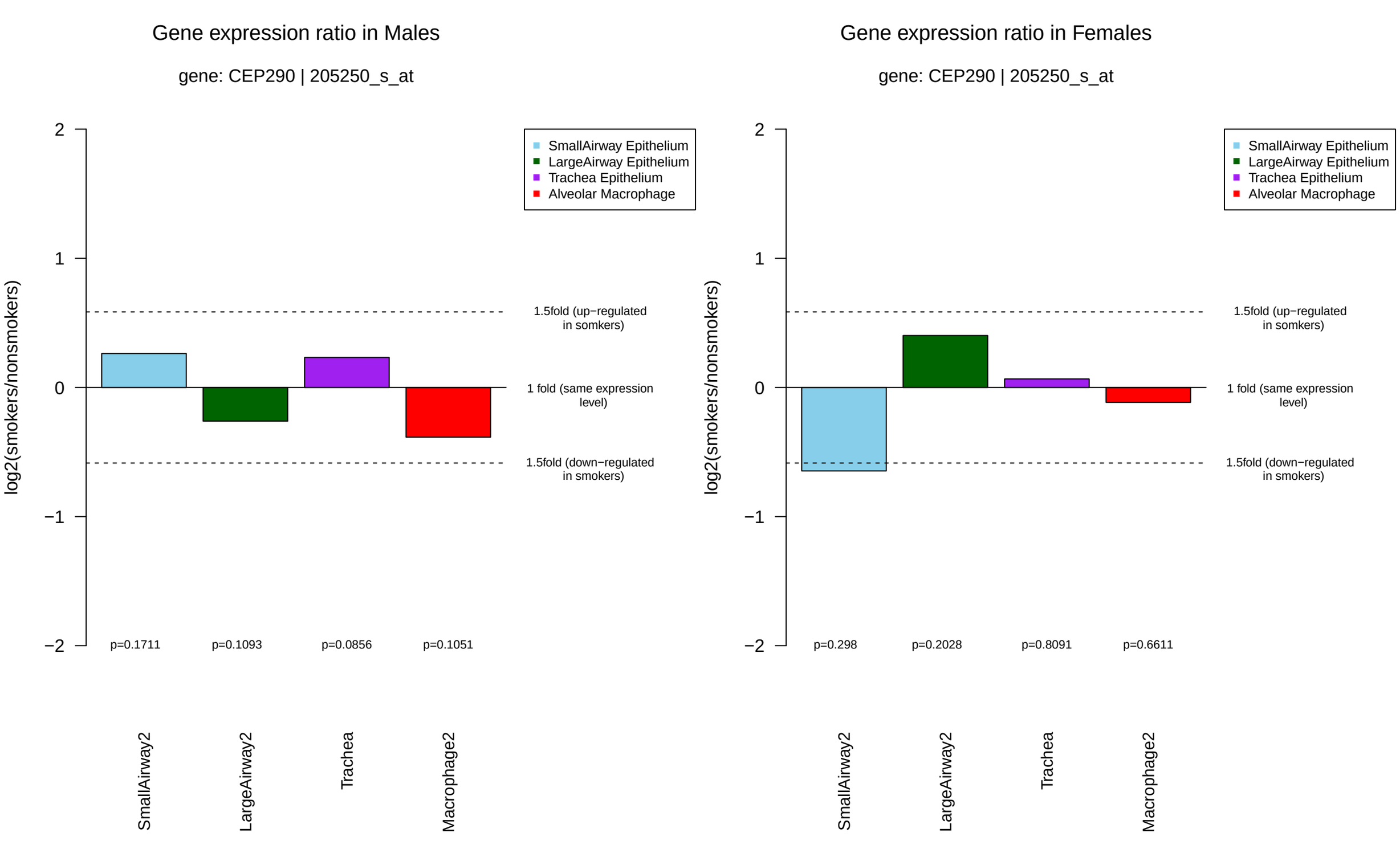
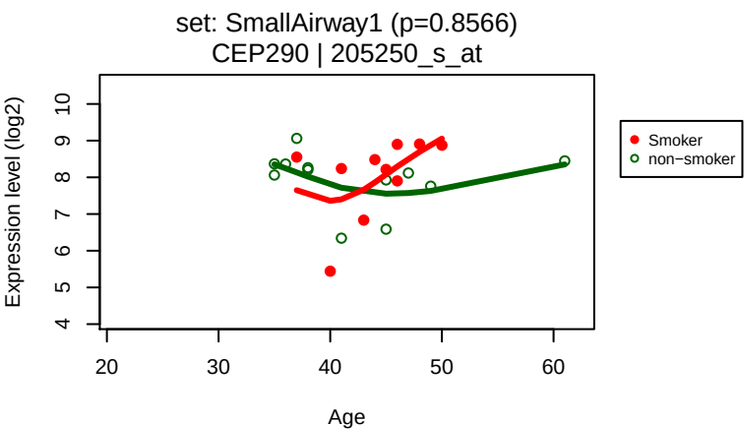
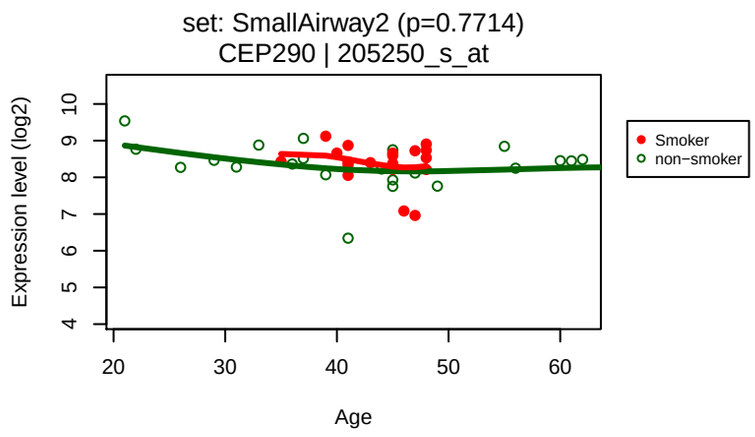
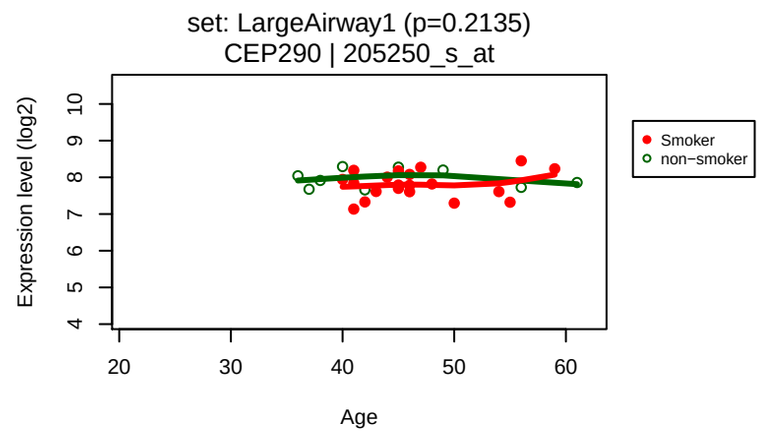

Gene expression of smokers and nonsmokers
gene: CEP290 | 205250_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8566
p=0.7714
p=0.5192
p=0.2135
p=0.858
p=0.0887
p=0.2697
p=0.0953
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

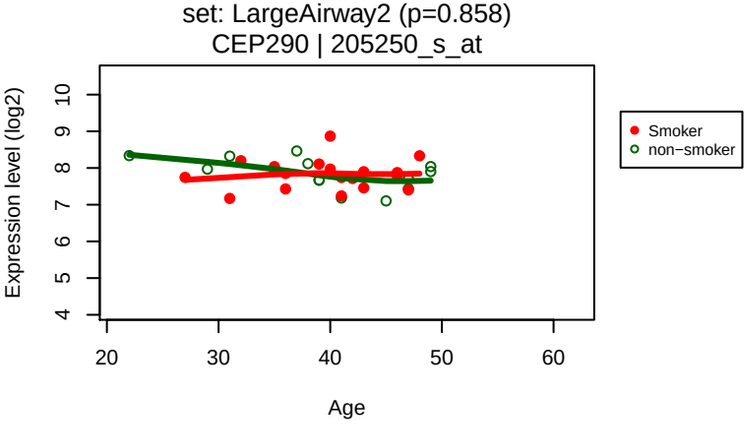

Gene expression ratio (smokers/nonsmokers)
gene: CEP290 | 205250_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8566
p=0.7714
p=0.5192
p=0.2135
p=0.858
p=0.0887
p=0.2697
p=0.0953
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

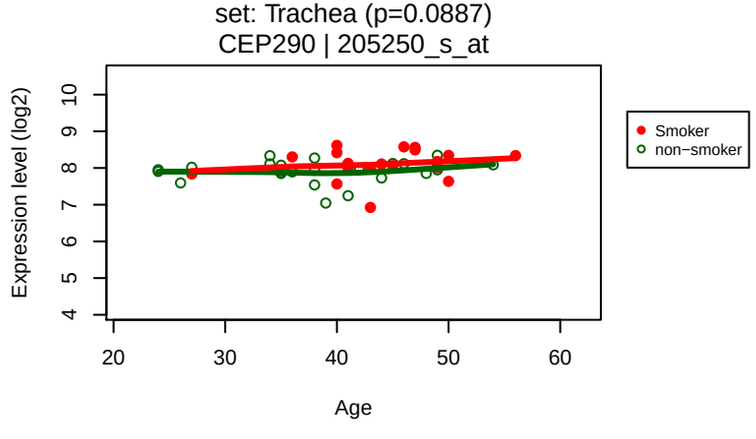

Gene expression ratio in Males
gene: CEP290 | 205250_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1711
p=0.1093
p=0.0856
p=0.1051
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEP290 | 205250_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.298
p=0.2028
p=0.8091
p=0.6611
SmallAirway2
LargeAirway2
Trachea
Macrophage2

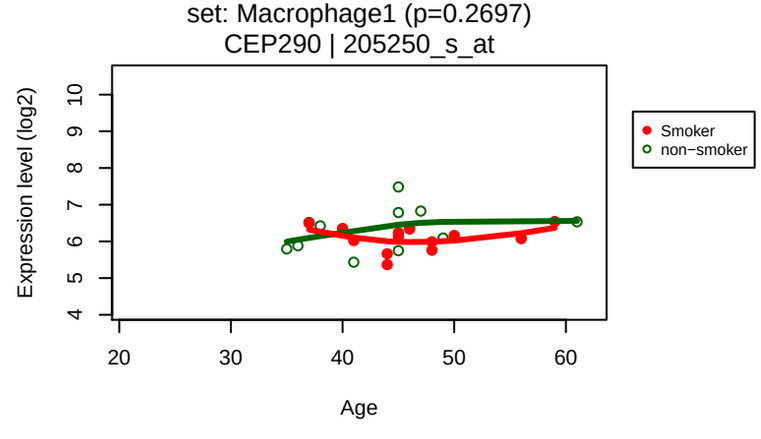

set: SmallAirway1 (p=0.8566)
CEP290 | 205250_s_at
Expression level (log2)
Age
Smoker
non−smoker

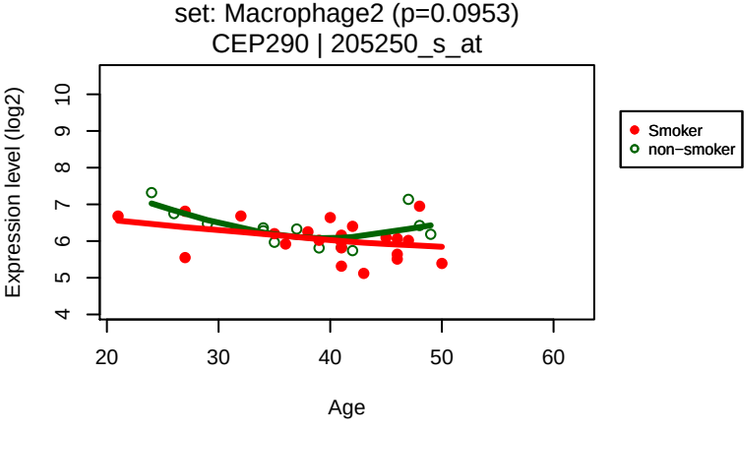

set: SmallAirway2 (p=0.7714)
CEP290 | 205250_s_at
Expression level (log2)
Age
Smoker
non−smoker

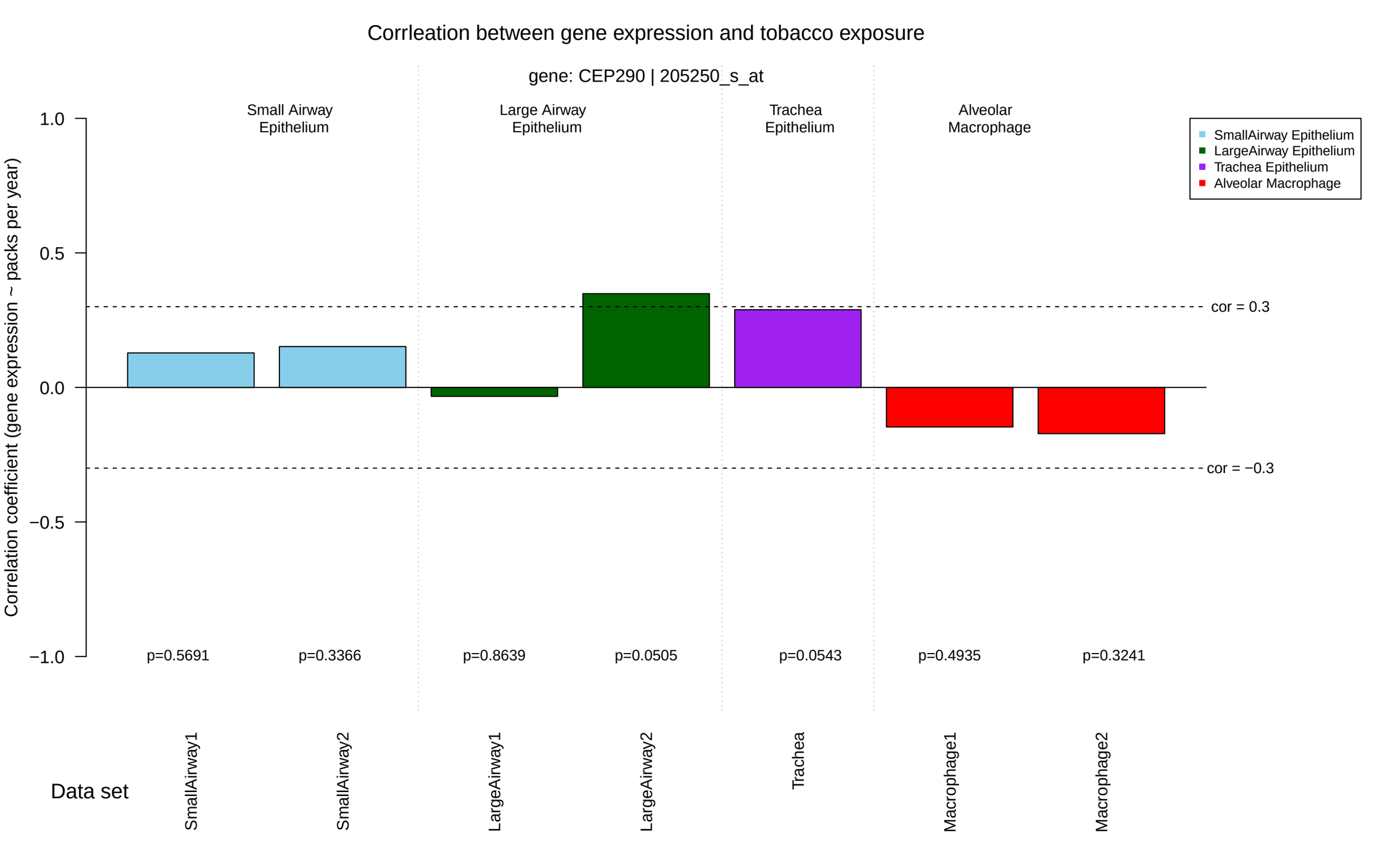

set: LargeAirway1 (p=0.2135)
CEP290 | 205250_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.858)
CEP290 | 205250_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0887)
CEP290 | 205250_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2697)
CEP290 | 205250_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0953)
CEP290 | 205250_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEP290 | 205250_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5691
p=0.3366
p=0.8639
p=0.0505
p=0.0543
p=0.4935
p=0.3241
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2